

Create a Dockerized Python or Node.js Application

Step 1 - Initialize a node application

- Run *npm init* to initialize a node app
- Run *npm i express* to install express

```
1RV24MC038_hrshreeraksha@asus:~$ mkdir lab3 && cd lab3
1RV24MC038_hrshreeraksha@asus:~/lab3$ npm init -y
Wrote to /home/1RV24MC038_hrshreeraksha/lab3/package.json:
```

```
{
  "name": "lab3",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
1RV24MC038_hrshreeraksha@asus:~/lab3$ npm i express
```

```
added 68 packages, and audited 69 packages in 16s
```

```
18 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
1RV24MC038_hrshreeraksha@asus:~/lab3$ ls
node_modules package.json package-lock.json
```

Step 2 - Create and write Dockerfile

- Run *touch Dockerfile*
- Insert the following commands into the docker file

```
1RV24MC038_hrshreeraksha@asus:~/lab3$ cat Dockerfile
FROM node:latest

WORKDIR /app

COPY package*.json
RUN npm install

COPY . .

EXPOSE 3000

CMD ["node", "server"]
```

Step 3 - Create and write server.js

- Run *touch server.js* and Insert the lines of code into the file

```
1RV24MC038_hrshreeraksha@asus:~/lab3$ cat server.js
const express = require("express")
const app = express()
const PORT = 3000

app.get("/", (req,res)=>{
    res.send("Output of LAB 3 - Dockerized node application")
})

app.listen(PORT, ()=>console.log("Listening on", PORT));
```

Step 4 - Build the Docker file

- Run the following command *sudo docker build -t lab3-node .*

```

1RV24MC038_hrshreeraksha@asus:~/lab3$ sudo docker build -t lab3-node .
[+] Building 113.0s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 153B
=> [internal] load metadata for docker.io/library/node:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:latest@sha256:7478f3725ef76ce6ba257a6818ea43c5eb7eb5bd424f0c3df3a80ff77203305e
=> => resolve docker.io/library/node:latest@sha256:7478f3725ef76ce6ba257a6818ea43c5eb7eb5bd424f0c3df3a80ff77203305e
=> => sha256:4a27c4980d1337802e96e45bef11ffee3cabee60674abbd7ebaf903f70c00560 2.49kB / 2.49kB
=> => sha256:708274aafe49b02dddc66f97a5c45bb0b8fcf481ce6b43785b11f287fd4e4e1b 48.48MB / 48.48MB
=> => sha256:078b2eece9b24f617524f986db4dd04f977e3e7d6fe15a9088a584147bc6ba05 64.40MB / 64.40MB
=> => sha256:7478f3725ef76ce6ba257a6818ea43c5eb7eb5bd424f0c3df3a80ff77203305e 5.14kB / 5.14kB
=> => sha256:9ebdc6b5e95f31f55f9d6f2010a57c4f20f1db8db1751e06ef0ed5bcd1ca1cb7 6.75kB / 6.75kB
=> => sha256:8cdf261ed5cee6fd4e729e68c2831a0abc6c7c017569ab45dfd2240bcc3712d 24.03MB / 24.03MB
=> => sha256:a1208d53eb0667932469017a5ef3960e5ed2aea9143d62b983abe2f8593eeb9a 211.46MB / 211.46MB
=> => extracting sha256:708274aafe49b02dddc66f97a5c45bb0b8fcf481ce6b43785b11f287fd4e4e1b
=> => sha256:78c780840163e6a715b547f995a5b90f747b89634f54deabd353c7ee16183496 3.33kB / 3.33kB
=> => sha256:435cbc1d6a03726bbc9f87d22a3b9d971e32104aafa770406a1231864e9e5b6b 56.16MB / 56.16MB
=> => extracting sha256:8cdf261ed5cee6fd4e729e68c2831a0abc6c7c017569ab45dfd2240bcc3712d
=> => sha256:29dd662fc17fdd9a1d4907978c65a9134787d96a9bf3b82e63dc7a7f21190c15 1.25MB / 1.25MB
=> => extracting sha256:078b2eece9b24f617524f986db4dd04f977e3e7d6fe15a9088a584147bc6ba05
=> => sha256:73d86704c819d129f8e98efcf40c6d3580c8a2f4f1a073f2856a8627275be80b 447B / 447B
=> => extracting sha256:a1208d53eb0667932469017a5ef3960e5ed2aea9143d62b983abe2f8593eeb9a
=> => extracting sha256:78c780840163e6a715b547f995a5b90f747b89634f54deabd353c7ee16183496
=> => extracting sha256:435cbc1d6a03726bbc9f87d22a3b9d971e32104aafa770406a1231864e9e5b6b
=> => extracting sha256:29dd662fc17fdd9a1d4907978c65a9134787d96a9bf3b82e63dc7a7f21190c15
=> => extracting sha256:73d86704c819d129f8e98efcf40c6d3580c8a2f4f1a073f2856a8627275be80b
=> [internal] load build context
=> => transferring context: 43.51kB
=> [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:845f070f4fe00dd5ffa8a924887a6b5681896a1d80deb65f28ea2ebd6ae329bc
=> => naming to docker.io/library/lab3-node

```

Step 5 - Run container

- Run the command `sudo docker run -p 3000:3000 lab3-node`

```

1RV24MC038_hrshreeraksha@asus:~/lab3$ sudo docker run -p 3000:3000 lab3-node
Listening on 3000

```

Output :

